

#HW 11

● Graded

Student

Anthony Escareno-Pinon

Total Points

100 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 1 pt for part b, the process can be possible as entropy transfer can be less than 0.
- 1 pt c, will be impossible. To make the entropy change 0 when the entropy transfer is more than 0, you need to have a negative entropy production, which is not possible.
- 1 pt d will be indeterminate because if entropy change is less than the entropy transfer then energy production can become negative which is not possible. Only when the entropy change is greater than or equal to the entropy transfer, the entropy production will be greater than or equal to 0 which makes the process possible.
- 1 pt g will be indeterminate too.
- 1 pt e will be possible.
- 5 pts It has not been explained whether the process will be possible or impossible or indeterminate.
- 1 pt f will be impossible since entropy production should always be positive.
- 1 pt a will be possible because in absence of entropy transfer, the entropy production must be positive to make entropy change greater than 0.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts Part c will be irreversible since, Q_h will be 1000 and as a result efficiency will be 60 percent.
- 2 pts Part a will be irreversible since efficiency will be 0.5, which is less than $2/3$.
- 2 pts b will be reversible since efficiency will be $2/3$. The work will be 600 kJ.
- 5 pts You have to mention which process is reversible, which is irreversible, and which is impossible.

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts For efficiency calculation, temperatures should always be converted to Kelvin or Rankine.
- 2 pts In part b, efficiency will be around 0.6. Answers will be different
- 1 pt the answer of c is not clear.
- 2 pts In part d T_h is 1000 and you need to find T_c .

Question 6

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The efficiency calculation is wrong.

Question assigned to the following page: [1](#)

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☐	6. The Second Law of Thermodynamics	11%	100%	
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CHE 201: Introduction to Thermodynamics

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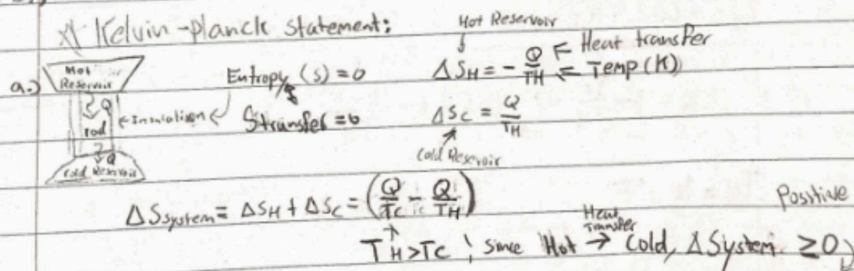
Questions assigned to the following page: [2](#) and [3](#)

CHE 201 HW 11

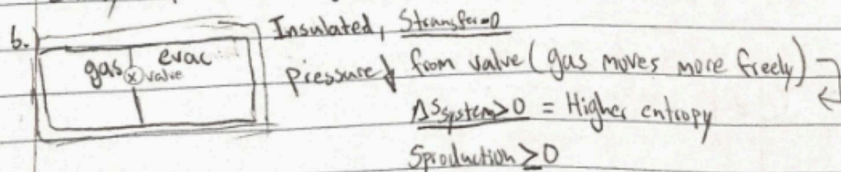
2.) 6.17.3 $\Delta S = S_{\text{transfer}} + S_{\text{production}}, S_{\text{production}} \geq 0$

	Entropy change	Entropy Transfer	Entropy Production
a.)	> 0	0	> 0 possible
b.)	< 0	< 0	> 0 {possible
c.)	0	> 0	Impossible (< 0)
d.)	> 0	> 0	0 or > 0 {Intermediate
e.)	0	< 0	> 0 {possible
f.)	> 0	Impossible (< 0)	< 0 {Impossible
g.)	< 0	< 0	0 {Intermediate

3.) 6.17.9



- Entropy is produced meaning Process is irreversible. $S_{\text{production}} = \text{positive} \Rightarrow \geq 0$



- Entropy is produced meaning Process is irreversible

Questions assigned to the following page: [4](#) and [5](#)

The %
Are
thermal
efficiency

$\eta < \eta_{\text{canot}} \rightarrow \text{Irreversible}$
 $\eta = \eta_{\text{canot}} \rightarrow \text{Reversible}$
 $\eta > \eta_{\text{canot}} \rightarrow \text{Impossible}$

1. $Q_H = Q_C + W_{\text{cycle}}$
2. $\eta = \frac{W}{Q_H}$
3. $1 - \frac{T_C}{T_H} = \eta_{\text{canot}}$

4.) 6.17.14 $\left\{ \eta_{\text{canot}} = 1 - \frac{T_C}{T_H} = 1 - \frac{400}{1200} = \frac{2}{3} \approx 66\% \right.$
 ≈ 0.5

(a) $\eta = \frac{450}{900} = 50\%$ Irreversible $\approx \frac{2}{3}$

(b) $W = Q_H - Q_C \Rightarrow 900 - 300 = 600$ $\eta = \frac{600}{900} = 66\%$ Reversible

(c) $Q_H = 600 + 400 = 1000 \Rightarrow \frac{600}{1000} = 60\%$ Irreversible

(d) $\eta = 75\% > 66\%$ Irreversible

5.) 6.17.17

(a) $1 - \frac{400K}{1600K} = \frac{3}{4} = 75\%$

(b) $1 - \frac{293.15K}{723.15K} = 0.62 \approx 62\%$ $0.62 = \frac{1000KJ}{Q_H}$

$Q_H = 1612.90 KJ$

$Q_C = 602.9 KJ$

$Q_H = \frac{1000KJ}{0.62} = 1612.90 KJ$

$Q_H - W_{\text{cycle}} = Q_C$

$\rightarrow 1612.90 KJ - 1000 KJ = 602.9 KJ$

(c) $0.6 = 1 - \frac{T_C}{T_H} \Rightarrow 0.6 = 1 - \frac{40^\circ F}{T_H} \rightarrow 0.6 - 1 = -\frac{40^\circ F}{T_H}$

$T_H = 100^\circ F$

$-0.4 = -\frac{40^\circ F}{T_H}$

(d) $0.4 = 1 - \frac{T_C}{T_H} \Rightarrow 0.4 - 1 = -\frac{T_C}{727^\circ C}$

$T_H = \frac{-40^\circ F}{-0.4}$

$(0.4 - 1) \cdot 727^\circ C = T_C$

$T_C = 436.2^\circ C$

Question assigned to the following page: [6](#)

6.) 6.12.19

(H) Earth Surf (1100°C) $\rightarrow 1373.15\text{K}$ ~

$$h = 1 - \frac{T_c}{T_H}$$

(C) Air (15°C) $\rightarrow 288.15\text{K}$ ↓

$$79\% \approx 0.79 = \left(1 - \frac{288.15\text{K}}{1373.15\text{K}}\right)$$

He is correct, the Claim Stands